

Use of mobile phone while studying

Alessio Gandelli¹

¹ *Università degli studi di Trento Italy*

Abstract—This paper focuses on the effects of the use of smartphones while studying. Through the collection of data from questionnaires, sensors, and time diaries, we studied correlations between personality traits and the use of notifications. Additionally, we explored general statistics on how students use their phones, their most used and installed apps, and top notifications. Our results demonstrated how messaging and social networking apps capture students' attention. Our findings can help inform users and educators on how to use smartphones in a more productive way.

Keywords—Human behaviour, smartphone usage, attention

I. INTRODUCTION

Nowadays, we have entered the era of the attention economy, where attention is regarded as a precious commodity. With such a vast array of possibilities, it can be difficult for people to focus on one thing for an extended period of time. Despite this, many students still claim that they can multitask without any negative consequences impacting their performance. However, research [1] paints a much different picture.

The truth is, due to the human brain's construction, it is virtually impossible for us to focus on more than one thing at a time. Rather, our attention continually shifts from one activity to the next, resulting in a considerable waste of resources [2]. Under these conditions, we must consider the concept of task switching and the costs it incurs. As Monsell [3] has pointed out, "subjects' responses are substantially slower and, usually, more error-prone immediately after a task switch". This phenomenon can have a significant impact on our ability to concentrate and focus on a single task.

In this research project, we will explore how the use of smartphones affects the quality of the students' studies. Through the use of sensors, surveys, and auto-reported time diaries, we will analyze the factors that lead people to use their phones while studying. In addition, we will study the role of notifications in the attention mechanism and whether there is a correlation with personality traits.

II. METHODS

In this section, we will explain all the steps required to perform the analysis, justifying the different choices: from data preprocessing to the analysis itself. This is a secondary data analysis, so it does not concern the data collection, which was done in the context of a broader project ¹.

We will briefly review the structure of the data and the transformations we made.

a. Data

In 2020, all University of Trento (UNITN) students were asked to participate in the data collection. The sample is composed of UNITN students, mainly Italians, Android users who accepted to participate in the study in exchange for a monetary compensation. More information about the data collection can be found here [4].

The data was collected from November 13, 2020 to December 12, 2020.

There are 3 sources of data, each of which generates a substantial amount of data. In this section, we will discuss only the subset we used.

- **Questionnaires:** The students involved completed a set of surveys to gather socio-demographic and personality (Big 5) data.
- **Sensors:** The iLog app installed on their phones collected data from 32 sensors, with touch of the screen, notifications and applications being the most relevant for this study.
- **Time diary:** The students who agreed to participate in the study were asked to download the iLog app, which was used to collect data from sensors, and to fill out a time diary every day for 29 days. The frequency of the time diary was every 30 minutes for the first two weeks, and every hour for the last two weeks.

Source	weight	explanation
Touch	6.0 GB	Every touch of the screen
Application	3.4 GB	The application in foreground
Notification	729 MB	Source of the notification
Diary	128 MB	What they were doing
Socio	111 KB	Data about the individual

TABLE 1: INITIAL DATASETS AND THEIR SIZE

For the purpose of this research, only users with complete data were considered. The population sample consisted of

¹<https://cordis.europa.eu/project/id/823783>

161 Italian students of the University of Trento, who were also Android users; 93 females and 69 males, of whom: 42 were enrolled in engineering courses, 33 in social sciences, 22 in law, 21 in natural sciences, 20 in economics, 14 in humanities, and 10 in other courses.

b. Preprocessing and cleaning

We prepared the data by following a process from raw data to the final dataset. The demographic and time diary data had already been cleaned, but the sensor data required our attention. Therefore, we cleaned it ourselves.

CSV to SQL The biggest datasets were all in CSV format. To manage them more efficiently, we first transformed the CSV files into a SQLite database. This allowed us to query only the useful data without loading it all into RAM. Additionally, it enabled us to store the data using a common schema.

First filtering The first exploration aimed at finding users with data from all sources: we used set operations to keep only users with all sensors, as seen in the figure 1. During this phase, all useless columns were also removed.

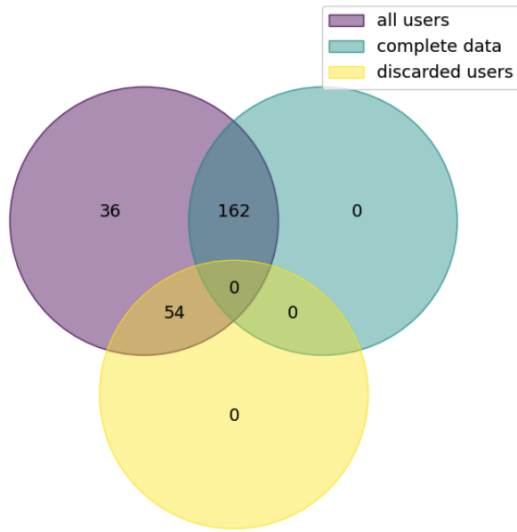


Fig. 1: Venn diagram of userid

Cleaning This phase is crucial for making the data more consistent and legible: it ensures uniformity and readability. These are the steps followed to clean the data:

- **App names:** The app names in both applications and notifications were the Android package name, which needed to be converted into a more readable name. The heuristic we used was to remove all occurrences of strings such as com, it, org, google, samsung, huawei, etc., and to keep only the last part of the string. For example, com.google.android.youtube became youtube. We also removed data that were linked to system logs or to iLog.

- **Age:** in the sociodemographic the age field was a cohort of ages, we converted it into a number.
- **Time diary** There were more than 30 possible values for the what field, we grouped them into few categories, the main are: sleeping, food, entertainment, work/class, other, missing.
- **User selection:** Only the 169 users with complete data were considered. Unfortunately, some users had missing data for certain sensors or had few entries in the time diary. As a result, we had to discard the 20 users with the most incomplete time diaries.

Yardstick The data was not consistent in terms of time for each sensor; in fact, we had many entries per minute (sometimes per second) for the touch sensor, while an entry every half an hour for the time diary. The goal of the yardstick is to have homogeneous data aggregated at the minute level, containing all the data that is useful for this analysis. Now, all the datasets can be fit into an 8 million row dataframe.

Study Time We extracted study minutes for each student. To ensure accuracy, we then proceeded to remove any outliers that had either 100% or 0% study time, as these could have skewed the results.

III. RESULTS

This section contains an analysis of the dataset created earlier. It begins with exploratory data analysis and then moves on to data related to the study.

a. Exploratory Data Analysis

Let us investigate general statistics on how students use their phones, their most popular and most installed apps, and the top notifications.

Application We can see from Fig 2 that the first four apps— WhatsApp, Instagram, YouTube, and Chrome— account for 50% of the usage. While the remaining 50% consists of a long tail of application used just few minutes by few students.

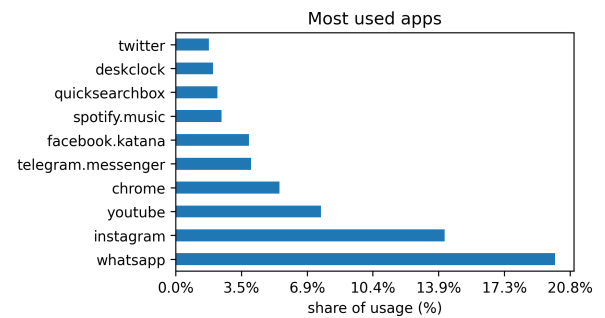


Fig. 2: Most used apps

By examining the most installed apps (Fig 3), we can observe that 97% of the students have downloaded Gmail. Although it is not widely used, it is essential to check emails nowadays, especially if the university email is a Google account.

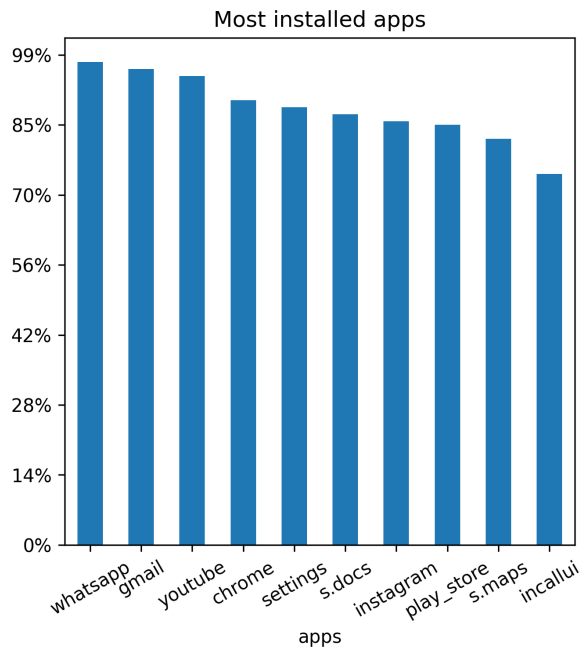


Fig. 3: Most used apps

Notification Notifications reveal the prevalence of WhatsApp, but Telegram is a close second despite not being among the most downloaded apps. It is particularly popular among students, who often join large groups and consequently receive numerous notifications.

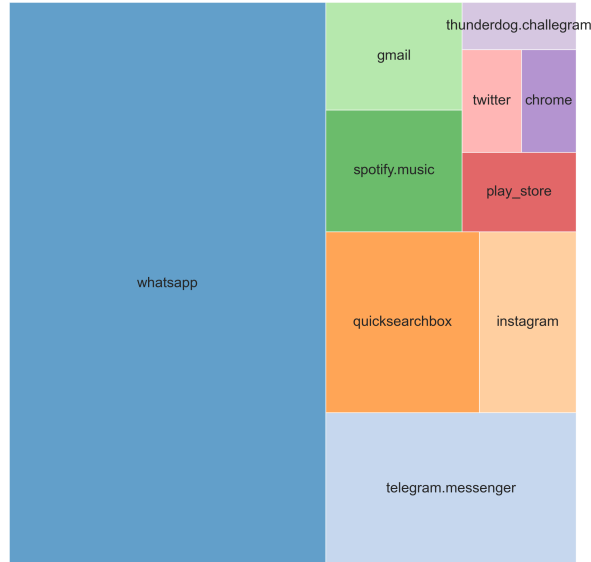


Fig. 4: Top notification

What people do Fig 5 presents an average day for the selected sample: we can clearly see a recurrent pattern in particular for sleep and food.

b. Study time

After exploring the data, we shifted the focus to the time students claimed they spent studying. We will analyze patterns of notifications, touches on the screen, and the most used applications.

We must take into account the period, as we are between

November and December. How people study is not homogeneous during the semester: we expect those with exams in December to study more than those with exams in January and February. Since we do not have this information, we must be careful in our interpretation of the results. If someone attending a specific course is studying more than the others, we cannot say that the course is more difficult or requires more effort. Rather, there may be different deadlines. For such an analysis, we need more longitudinal data.

From the data we have we can try to answer some questions, such as:

- How much time do they study?
- How much time passes between two notifications?
- How much time passes between two phone sessions?
- How frequently students touch their phone?
- What apps are they using?

Activity	daily share
Sleeping	0.30
Entertainment	0.14
Study 2	0.11
Food-related	0.07
work/class	0.05

TABLE 2: SHARE OF THE MAIN ACTIVITY

According to Fig 6 students reported that on average they study 3.9 hours a day with a std of 1.6.

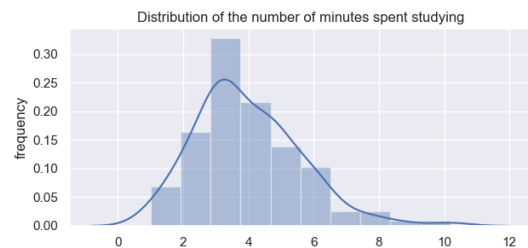


Fig. 6: Duration of study session in hours

Notification Fig. 7 shows the distribution of the average number of minutes between notifications. The top one considers all the samples, but since it has a long tail, it is useful to consider only the 90th percentile, i.e. removing the last 10% of observations. The median is around 6, which means that, on average, half of the students receives at least one notification every 6 minutes. This could potentially be a source of distraction. We cannot know if the mobile phone was in another room or in their pocket, but we can look at the number of touches of the phone in the same time span. Even though a notification is not necessarily a source of distraction, a lack of touch does not mean that the student has not been distracted by the phone. The results in table 5 show a moderate positive correlation of 0.53 with a p-value of 10e-10, which is highly significant. From this result, we can conclude that the more notifications a person receives, the more likely they are to move their focus to the phone.

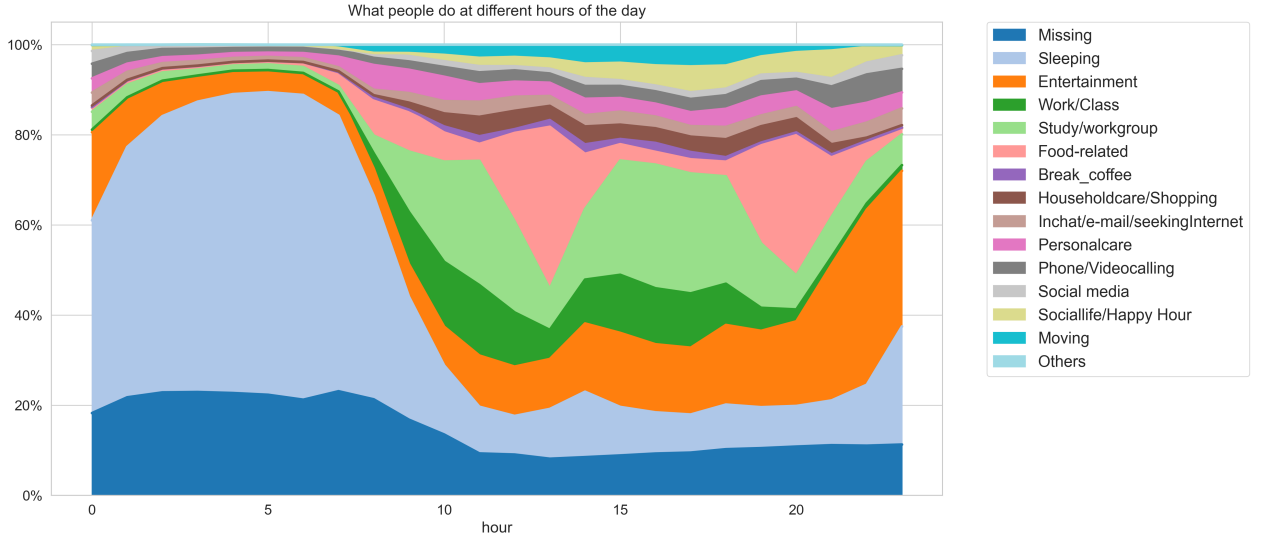


Fig. 5: an average day

	average	median	std	max streak
count	142	142	142	142
mean	108.7	75.2	128.5	1687.4
std	780.4	767.9	242.0	2743.3
25%	15.5	3.0	62.0	640.0
50%	20.3	4.0	70.0	770.0
75%	28.7	5.7	93.5	1312.0

TABLE 3: NOPHONE

Pauses from phone At this point, someone could argue that the touches could be all concentrated in a texting session. To address this, we conducted a more sophisticated analysis of the touch data. The goal was to look at the pauses from the phone between sessions of usage. For each student, we have a list where each value is the number of consecutive minutes that the student did not touch the phone. For example, a list [12, 6, 2, ..., 2, 3] means that this student did not use the phone for 12 minutes, then touched it for a while, then no phone for 6 minutes, and so on.

In other words, for each student, we have a list with a length equal to the number of times they put the phone in stand-by, and the value is the number of minutes before the next pick-up.

Then, from this vector, we computed some statistics for each user, such as: mean, median, standard deviation, and maximum streak without the phone. The results are in minutes and the grouped results are in Table 3, which shows the same skewness as the notification analysis. This table can be confusing, as we are computing the statistics of the statistics. Each student has their own average minutes without phone; here, we are doing the mean of this value for all the sample. The mean is 108 minutes, which is a high value due to a few outliers. If we look at the 75 percentile of the median, we can see that for 3/4 of the students, most of their pauses from the phone are less than 6 minutes long.

Apps used Students often claim to be studying, but let's take a look at the apps they use. To avoid outliers, I counted the top three apps used by each student. The results are shown in Table 4. Whatsapp was the most popular, with 74%

of students having it in their top three. Instagram was second, with 53%, and Telegram was third, with 17%. These results demonstrate how messaging and social networking apps capture students' attention.

	app name	share of user
0	whatsapp	.74
1	instagram	.53
2	telegram.messenger	.17
3	youtube	.15
4	chrome	.15
5	facebook.katana	.13
6	spotify.music	.08
7	incallui	.07
8	twitter	.05
9	calendar	.03

TABLE 4: TOP USED APPS WHILE STUDYING

Correlation with personality In Table 5, we can observe the correlations between minutes of studying, touches to the screen, and notifications received, as well as their correlations with personality traits. The results for personality traits were surprising, as we expected a correlation between neuroticism and inappropriate use of the phone while studying. However, the only statistically relevant value was a low negative correlation between openness and touches.

We must also take into account the restricted sample and the fact that the precision of the data is questionable, as many simplifications have been made up to this point.

IV. CONCLUSION AND DISCUSSION

This research project investigated how the use of smartphones influences the study quality. Surveys, auto-reported time diaries, and sensor technology were used to evaluate the correlation between personality traits and the use of notifications. Additionally, the impact of notifications on the attention mechanism was studied to answer questions raised in the introduction, and gain insight into the true effects of smartphones on students' studies.

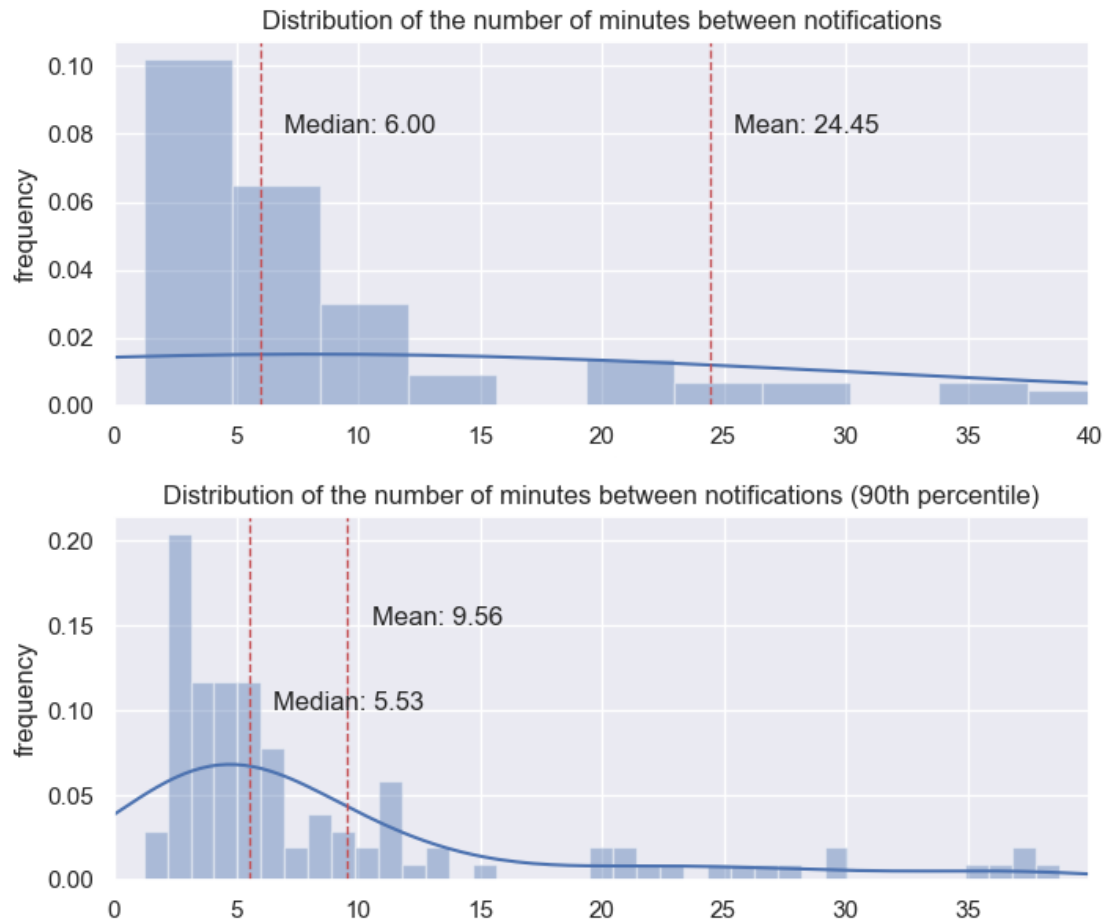


Fig. 7: Time between notifications

	minutes	touches	notification
minutes	1.0***	0.592***	0.342***
touches	0.592***	1.0***	0.531***
notification	0.342***	0.531***	1.0***
Extraversion	0.058	0.096	0.089
Agreeableness	0.033	0.045	-0.003
Conscientiousness	0.094	-0.009	0.026
Neuroticism	-0.053	-0.002	-0.001
Openness	-0.026	-0.186*	-0.142

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

TABLE 5: CORRELATION

The results of the analysis revealed that the first four apps - WhatsApp, Instagram, YouTube, and Chrome - accounted for 50% of the usage, with a long tail of other apps. Notifications revealed the prevalence of WhatsApp, but Telegram was a close second despite not being among the most downloaded apps. While studying, students often pick up their phones, both due to notifications and an internal trigger that draws their attention to the phone.

The most used apps are related to communication with others. We live in a society that is somewhere between hyper-connected and lonely. The fear of missing out dominates our lives: when a notification arrives, we feel the need to check what is happening. The structure of these applications does not help. People often look at screen time as a metric to define the attention given to the phone, but we argue that

the number of times the phone steals our attention is more relevant, since context switching is not an easy task for our brains, and we lose a lot of efficiency when multitasking.

Overall, this study showed that smartphones have a significant impact on students' study. It highlighted the relevance of notifications, as well as the need to limit distractions while studying. In conclusion, it is essential to be aware of how phone use can affect our concentration and focus, and to use our phones more consciously.

REFERENCES

- [1] K. Mokhtari, J. Delello, and C. Reichard, "Connected yet distracted: Multitasking among college students," *Journal of College Reading and Learning*, vol. 45, no. 2, pp. 164–180, 2015.
- [2] P. A. Kirschner and P. De Bruyckere, "The myths of the digital native and the multitasker," *Teaching and Teacher Education*, vol. 67, pp. 135–142, 2017. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0742051X16306692>
- [3] S. Monsell, "Task switching," *Trends in Cognitive Sciences*, vol. 7, no. 3, pp. 134–140, 2003. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1364661303000287>
- [4] M. B. R. C.-A. M. R. M. Z. C. G. G. V. A. D. G. P. K. A. G. A. C. G. G. S. S. M. B. L. C. A. H. J. L. Z. Fausto Giunchiglia, Ivano Bison, "A worldwide diversity pilot on daily routines and social practices (2020)," *University of Trento Technical Report. No. DISI-2001-DS-01*, (2021), 2020.